

Question:

What is Patient A's primary documented cancer diagnosis?

GraphRAG + Neo4j:

Based on the provided records, the patient's documented cancer diagnoses include "Breast Cancer" , "Metastatic Breast Cancer" , "Metastatic Malignant Breast Cancer" , "Carcinoma of right breast" , "Metastatic Carcinoma Breast" , "Carcinoma Breast" , "CA BREAST" , and "Metastatic carcinoma right breast" .

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Based on the provided medical records, the patient's primary documented cancer diagnosis is breast cancer (specifically diagnosed in 2004 as breast T3N0M0 adenocarcinoma) .

Question:

What are Patient A's ER, PR, and HER2 statuses?

GraphRAG + Neo4j:

Based on the available records, the patient's biomarker statuses are documented as follows:

ER (Estrogen Receptor): Positive (+) on 2004-06-29 and 8/8 on 2017-01-02 .

PR (Progesterone Receptor): 0/8 on 2004-06-29 and 0 on 2017-01-02 .

HER2: Documented as Human Epidermal Growth Factor Receptor 2 on 2004-06-29 and 1+ on 2017-01-02 .

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Based on the provided records, the patient's hormonal profile is as follows:

Estrogen Receptor (ER): positive (8/8)

Progesterone Receptor (PR): negative (0)

HER2 (Her 2 neu) status: negative (1+)

Question:

What chemotherapy or cancer treatments has Patient A received?

GraphRAG + Neo4j:

Based on the provided records, the Patient has received the following chemotherapy and cancer treatments:

Medications administered:

Paclitaxel (on 2004-06-29 and Cycle c6 on 2017-05-12 )

Anastrozole (on 2004-06-29)

Zoledronate (on 2004-06-29)

Palbociclib (on 2004-06-29)

Trastuzumab (Cycle c6 on 2017-05-12)

Regimens administered:

Neoadjuvant Chemotherapy (on 2004-06-29)

Epirubicin + Methotrexate + 5-Fluorouracil (on 2004-06-29)

Epirubicin + Cyclophosphamide (on 2004-06-29)

Procedures performed:

Radiotherapy (on 2004-06-29)

Modified Radical Mastectomy (on 2004-06-29)

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Based on the provided records, the patient has received the following chemotherapy and immunotherapy drugs and treatments as part of the protocol "Taxol D1,D8,D15 + Herceptin wkly c6" for a diagnosis of CA BREAST:

Inj Taxol (administered in 500 ml NS, 3-hour infusion)

Herceptin

Premedications given include: Inj Ondansetron, Inj Dexamethasone, Inj Ranitidine, and Avil .

Question:

What metastatic sites are documented for the Patient?

GraphRAG + Neo4j:

Not documented in the available records.

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Based on the provided records, the documented metastatic sites include the costal/mediastinal, diaphragmatic surfaces, and fissural right pleura (multiple nodular pleural deposits), as well as a deposit along the mediastinal pleura showing loss of the fat plane with the pericardium at places .

Question:

Give me a chronological summary of Patient's cancer history based on the available records.

GraphRAG + Neo4j:

Based on the available records, here is the chronological summary of the patient's cancer history:

2004-06-29: Diagnoses of Breast Cancer, Metastatic Breast Cancer, and Metastatic Malignant Breast Cancer were documented in the oncology flowsheet .

2016-12-02: A radiology PET-CT documented diagnoses of Carcinoma of right breast and Metastatic Malignancy .

2016-12-08: Right pleural thickening was documented in the oncology flowsheet .

2016-12-13: Cytopathology documented a diagnosis of Metastatic Carcinoma Breast .

2016-12-19: Histopathology documented Reactive lymphoid tissue .

2017-01-02: Histopathology documented diagnoses of Metastatic Carcinoma and Carcinoma Breast .

2017-05-12: A diagnosis of CA BREAST was documented in the chemo drug administration record .

2017-12-28: A radiology PET-CT documented a diagnosis of Metastatic carcinoma right breast .

2018-03-20: A radiology CECT documented a diagnosis of Metastatic Breast Cancer .

Undated: A diagnosis of "Metastasis" was documented in a general lab report .

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Based on the available medical records, the chronological summary of the patient's cancer history is as follows:

2004: Diagnosed with right breast cancer (T3N0M0) at SN Medical College, Agra . She presented with a lump in the right breast present for 3 years, identified as an adenocarcinoma .

2004 (Treatments): Received 4 cycles of IMF chemotherapy at SNMC-Agra , underwent a Modified Radical Mastectomy (MRM) , and External Beam Radiation Therapy (EBRT, 25 fractions/40.2 Gy) from August 25, 2004, to December 3, 2004 . She was also on Tamoxifen for 6 months .

2010: Remained asymptomatic until 2010 .

December 2016: Presented with recurrent/metastatic disease symptoms (including pain, dyspnea, and right-sided pleural effusion) . A PET-CT scan on December 2–3, 2016, showed evidence of metastatic disease involving the mediastinum, lymph nodes (left intra-clavicular, anterior diaphragmatic, mediastinal, bilateral hilar, and left external iliac), pancreas head, lungs (pleural deposits), and skeleton (lytic/sclerotic bone lesions in the parietal bone, left femur, vertebrae, and sacrum) . Pleural fluid cytology confirmed metastatic carcinoma .

Late 2016 to 2018: Underwent further evaluations and management plans at Dr. B.R.A. Institute Rotary Cancer Hospital (AIIMS, New Delhi), which included chemotherapy (such as Paclitaxel and EC regimens), targeted/hormonal therapies (such as Letrozole, Anastrozole, and Palbociclib), and management of complications like SVC thrombosis and pleural effusion .

May 2018: Noted with complaints of headaches and temporal/cerebellar lesions on imaging .

Question:

What did the pleural-fluid cytology report show for the Patient?

GraphRAG + Neo4j:

Not documented in the available records.

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The pleural-fluid cytology report showed lymphocytes, reactive mesothelial cells, and a few clusters of atypical epithelial cells, which are consistent with metastatic carcinoma breast .

Question:

Is a TNM stage explicitly documented to Patient A?

GraphRAG + Neo4j:

Yes, a TNM stage is explicitly documented as T:T3 N:N0 M:M0 on 2004-06-29 .

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Yes, a TNM stage of T3N0M0 is explicitly documented .

Question:

What type of breast carcinoma was diagnosed in Patient B, and what was its histological grade?

GraphRAG + Neo4j:

Not documented in the available records.

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The patient was diagnosed with invasive duct carcinoma, NOS (Not Other Specified), and its histological grade was Grade II (Nottingham's modification of Bloom Richardson score) .

Question:

What was Patient B's preoperative stage and what pathological stage was reported after surgery?

GraphRAG + Neo4j:

Not documented in the available records.

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Patient B's preoperative stage is not documented in the available records . The pathological stage reported after surgery is also not documented in the available records, though the histopathology report notes features of invasive duct carcinoma, NOS, Grade II .

Question:

What surgery did the Patient undergo, and was axillary clearance performed?

GraphRAG + Neo4j:

Based on the provided records, the patient underwent a lumpectomy on 2014-01-08 and a Modified Radical Mastectomy (MRM) / Left mastectomy on 2014-05-28 . The records also indicate that the patient underwent a "lymphnode dissection" , but whether an axillary clearance was specifically performed is not explicitly documented in the available records.

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The patient underwent a Left Modified Radical Mastectomy (MRM) . Yes, axillary clearance was performed up to level II .

Question:

What chemotherapy did Patient B receive in 2018, including the documented drugs and doses?

GraphRAG + Neo4j:

Not documented in the available records.

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In 2018 (specifically on 04/15/2018), Patient B received chemotherapy consisting of Inj. Epirubicin (administered in 110 ml NS over 1 hour) and Inj. Endoxan (Cyclophosphamide, 750 mg administered in 500 ml 5% D over 2 hours) .

Question:

What did the later pathology report show about Patient B's recurrent carcinoma, including ER, PR, HER2 and Ki-67?

GraphRAG + Neo4j:

Based on the pathology report dated January 10, 2018, Patient B's results showed the following:

ER: positive (8/8)

PR: positive (8/8)

HER2: 3+ and Ki67: 12-15%

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The later pathology report showed features of an invasive carcinoma, NST, with an impression of recurrent carcinoma . The hormonal profile of the tumor is ER-positive (8/8) (Allred score), PR-positive (8/8), and Her 2 neu-3+, with a Ki-67 proliferative index of 12-15% .

Question:

How did Patient B's breast cancer history evolve from the initial 2014 diagnosis to the later recurrence?

GraphRAG + Neo4j:

Not documented in the available records.

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Patient B's breast cancer history evolved as follows:

Initial Diagnosis (2014): In early 2014, the patient was diagnosed with infiltrating duct carcinoma (IDC) of the left breast (stage rT2N1Mo) . Her initial treatment included a private lumpectomy on January 14, 2014, followed by a left modified radical mastectomy (MRM) with axillary clearance on May 28, 2014 . Histopathology from the initial presentation showed the tumor was Estrogen Receptor (ER) positive, Progesterone Receptor (PR) positive, and Her2/neu positive . She defaulted on planned adjuvant hormonal therapy .

Recurrence (2018): Approximately 4 to 5 years later, the patient presented with a local recurrence manifesting as a swelling and nodule on the chest wall/scar site . In January 2018, she underwent an excision of the left chest wall nodule . Pathology of the recurrent lesion (measuring 2.3 x 1.8 x 0.6 cm) demonstrated an invasive carcinoma, NST (No Special Type), which maintained a positive hormonal and receptor profile (ER-positive, PR-positive, and Her2/neu 3+) with a Ki-67 proliferative index of 12-15% . Subsequent management for the recurrence included chemotherapy and radiation therapy plans .

Question:

Compare the ER, PR and HER2 status of Patient A and Patient B.

GraphRAG + Neo4j:

Patient A

ER Status: Documented as positive (+) on 2004-06-29 and 8/8 on 2017-01-02 .

PR Status: Documented as 0/8 on 2004-06-29 and 0 on 2017-01-02 .

HER2 Status: Documented as Human Epidermal Growth Factor Receptor 2 on 2004-06-29 and 1+ on 2017-01-02 .

Patient B

ER Status: Documented as positive 8/8 (undated) , positive (+) on 2014-05-28 , 8/8 on 2014-05-28 , and positive (8/8) on 2018-01-10 .

PR Status: Documented as positive 8/8 (undated) , positive (+) on 2014-05-28 , 78 on 2014-05-28 , 8/8 on 2014-05-28 , and positive (8/8) on 2018-01-10 .

HER2 Status: Documented as positive 3+ (undated) , positive (+) on 2014-05-28 , 3+ on 2014-05-28 , and 3+ on 2018-01-10 .

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For Patient A, the ER (estrogen receptor) is receptor-positive (8/8), the PR (progesterone receptor) is receptor-negative (0), and the HER2 (Her 2 neu) status is negative (1+) .

Question:

Which patients have documented metastatic disease, and what metastatic sites are documented for each?

GraphRAG + Neo4j:

Based on the provided medical records, here is the documentation regarding metastatic disease for each patient:

Patient A

Documented Metastatic Disease: Yes (diagnoses include "Metastasis", "Metastatic Breast Cancer", "Metastatic Malignant Breast Cancer", "Metastatic Malignancy", "Metastatic Carcinoma Breast", "Metastatic Carcinoma", and "Metastatic carcinoma right breast") .

Documented Metastatic Sites: Not explicitly documented in the available records (other than reference to the right breast primary/carcinoma) .

Patient B

Documented Metastatic Disease: Yes (diagnoses include "Metastatic" and "Skeletal metastasis") .

Documented Metastatic Sites: Skeletal system ("Skeletal metastasis") .

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Based on the provided medical records, here is the documentation regarding metastatic disease for each patient:

Patient A: Has documented metastatic disease with the following metastatic sites:

Pleura (multiple nodular pleural deposits on costal, mediastinal, diaphragmatic surfaces, and fissural right pleura) .

Lungs (two tiny subpleural nodules in the lingula and segment of the left lower lobe) .

Lymph nodes (enlarged prevascular and retrosternal lymph nodes) .

Bones (lytic sclerotic changes with associated soft tissue) .

Patient B: Has documented metastatic disease with the following metastatic site:

Skeleton (skeletal metastasis, noted as evidence of skeletal metastasis on the bone scan) .